

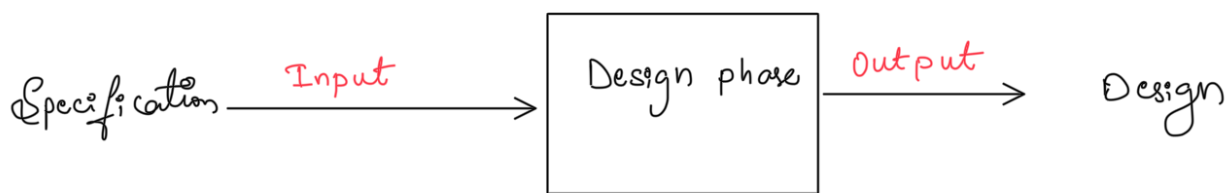
Design phase of Software Development

Day 101
19/10/2025

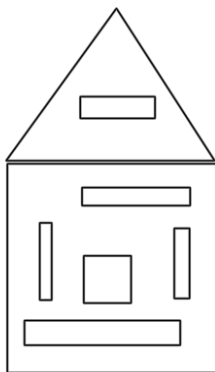
Design phase is one of the **most important** phase of Software Development

Generally the Design phase is lead By **Senior Engineers** or **leads** in the team

The software Specification is the input for the Design phase and the output is the Design



like **planning** a House Before **constructing**



structural Design

How many pillars it have, How beams it have and How would be the design of the plain and what would be its load bearing capacity.

Interior Design, front elevation design

Even for a house like have various designs

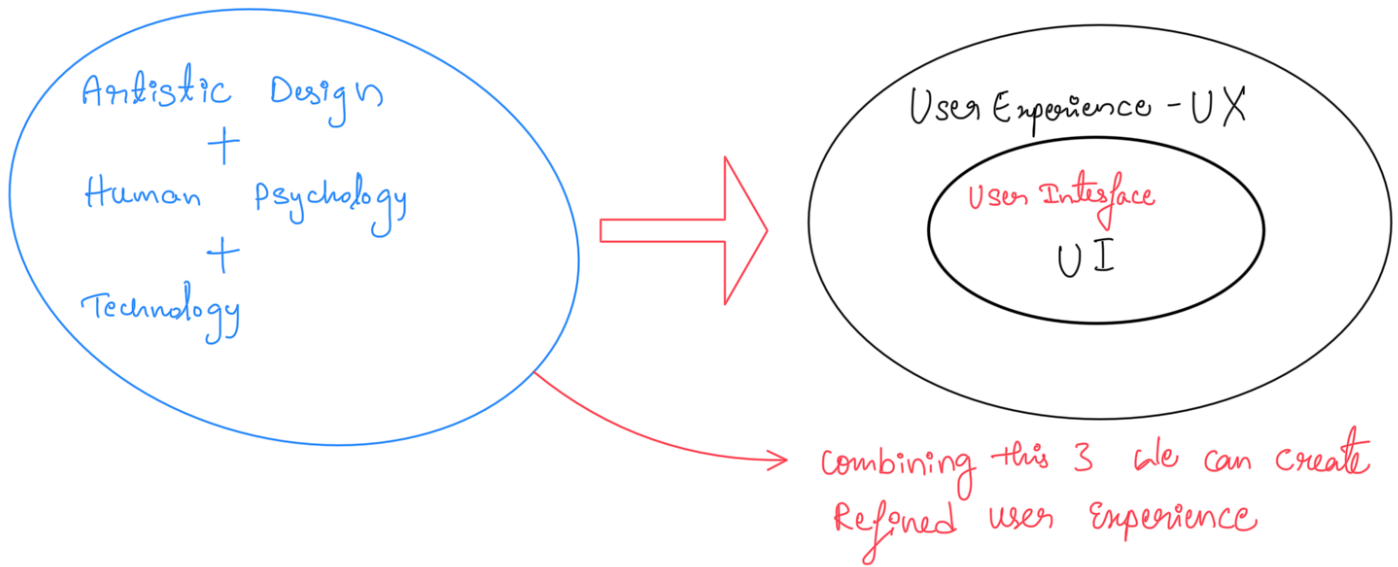
Similarly for a Software also we have various designs



UI Designs (It's Not about coloring, it's about User Experience improvisation)

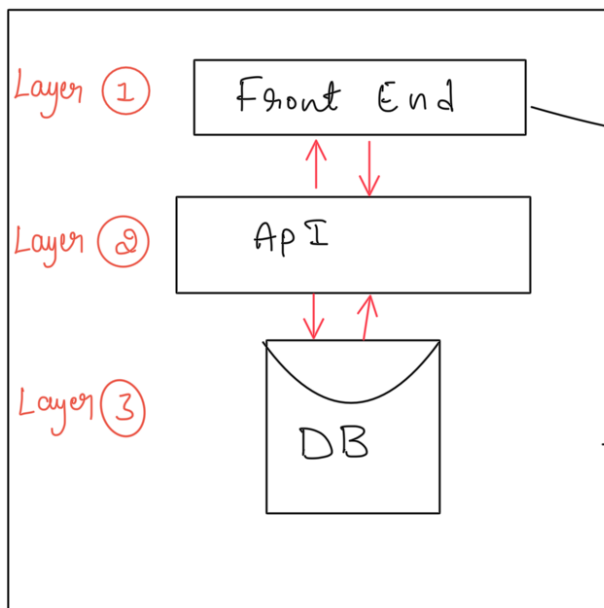
making them look Beautiful, easy to use, shiny Become essential.

It's Not Just About Coloring



System Design for a Simple Survey Web Application

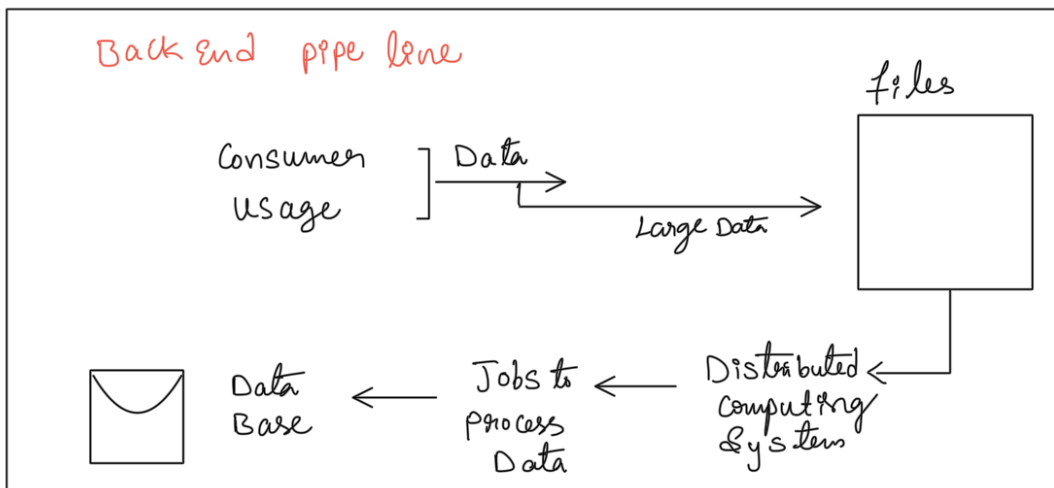
High Level Design



Web App Which should Support Desktop, laptop, tablet, Mobile

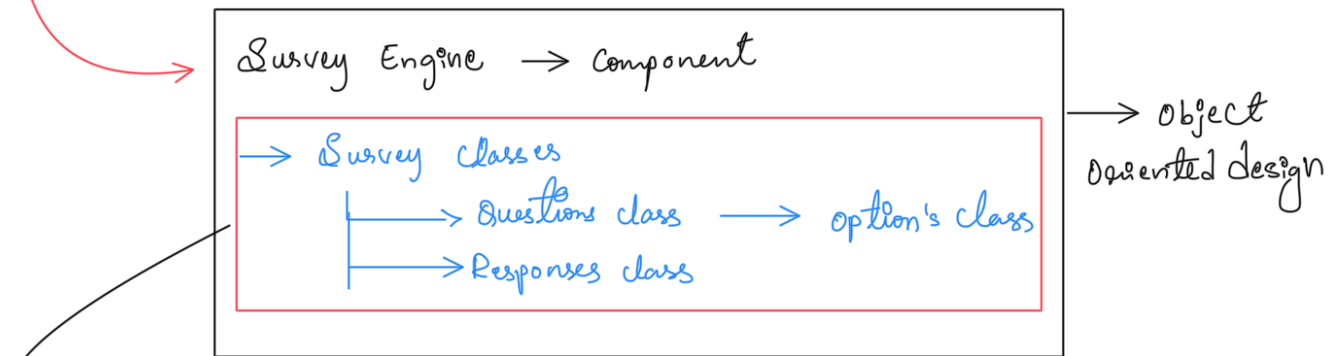
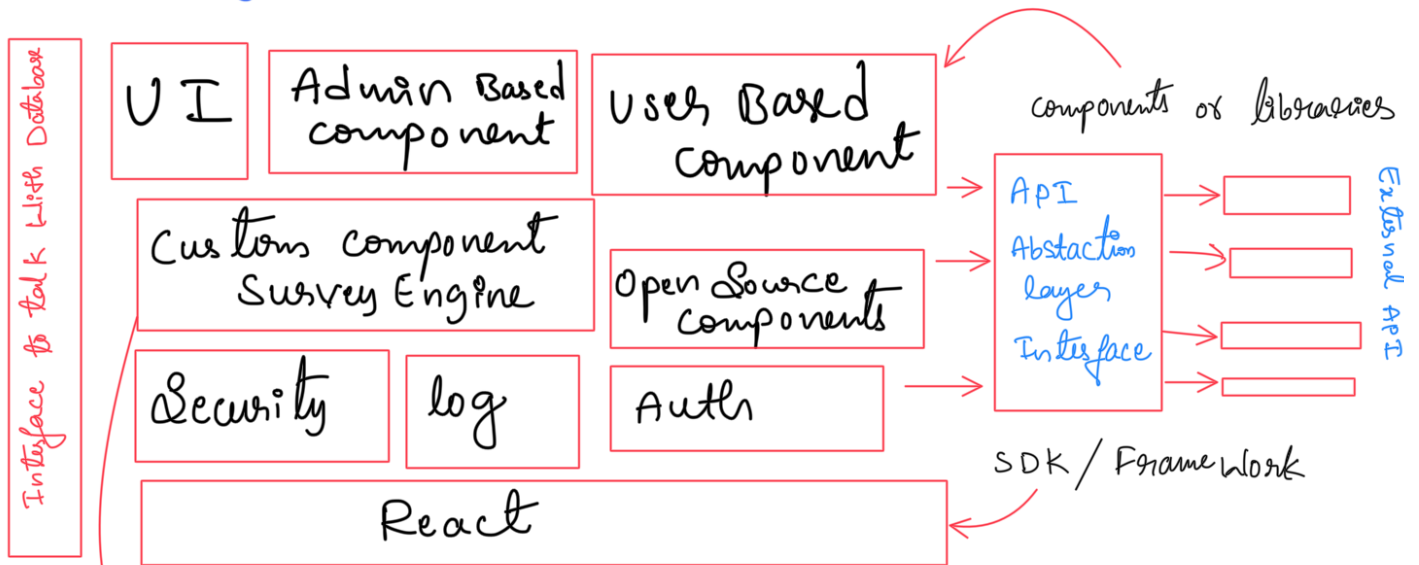
Have to Support this Various Devices

3 Layer or 3 Tiers Architecture
Entire Software is Broken Down into 3 layers each one of them are Capable of Running Independently.



Not in Real time But With increasing Consumers Data

Low Level Design



These classes perform some activities
 activities turned into function
 We can specify what will be the input and output to that function



API Also have component design



NOTE ; Frameworks are the Reusable code that can be used in Software development and that can reduce the time and Energy of Writing code from scratch.

Data Base Design

High level Design

choose type of Data Base SQL or NoSQL

How Many DataBases used Single or Multiple Data Base

Data partitioning, Load Balancing, scalability.

Low Level Design

Tables

- Name of Table's
- Columns of Table's
- Data type of Table's
- primary key, Foreign key
- How to Normalize Data, DeNormalization etc

single person can't write these all components code
obviously we need a Team to Build the Respective components
and Integrate them test debug and Deploy.

